OIL vs S&P 500

EVENT ANALYZER

Geopolitical Shocks & Market Impact Analysis

Nidhi Tangam Dayananda

Data Coverage

December 2014 – March 2026 | 2,828 Trading Days

Source: Bloomberg Terminal (Part A) / Yahoo Finance via yfinance (Part B)

Abstract

This project analyses the dynamic relationship between crude oil prices and the S&P 500 equity index across 2,828 trading days from December 2014 to March 2026. Using a modular Python package built from scratch, we examine eight major geopolitical and macroeconomic event windows and extend the analysis to 26 individual S&P 500 stocks across six sectors using a multiple regression model.

Our central finding is that the oil–equity relationship is not stable. Across the full dataset, oil explains only 5.4% of daily S&P 500 variation ($R^2 = 0.054$). Yet this average masks dramatic shifts during crises — particularly the 2025–2026 US-Israel-Iran war, during which the correlation turned sharply negative (−0.50) as oil surged +41.4% while the S&P 500 fell −5.2%.

The multi-stock extension reveals that this aggregate figure conceals large sector-level differences. Energy stocks carry strongly positive oil sensitivity while airlines are negatively exposed. Critically, a supply-driven oil shock (geopolitical) and a demand-driven shock (recession) produce opposite cross-sectional return patterns, demonstrating that the direction of oil's causality matters as much as its magnitude.

1 Executive Summary

This report presents findings from a Python-based analysis of the dynamic relationship between crude oil prices and equity markets from December 2014 to March 2026. The analysis is structured in two connected parts.

Part A analyses the relationship between oil and the broad S&P 500 index using a simple OLS regression across 2,828 trading days and eight major event windows. The baseline model confirms that oil explains only 5.4% of daily S&P 500 variation on average, while demonstrating that this relationship reverses sharply during geopolitical supply shocks.

Part B extends the analysis to 26 individual S&P 500 stocks across six sectors using a multiple regression model that controls broad market returns and the VIX volatility index, isolating the pure oil sensitivity of each stock.

Key Finding

Oil and the S&P 500 are weakly correlated on average ($\beta = 0.091$, $R^2 = 0.054$), but geopolitical shocks fundamentally alter — and often reverse — this relationship. The multi-stock extension reveals that Energy stocks and Airlines move in opposite directions during supply shocks, a pattern invisible at the index level.
Iran War 2025–26: Oil +41.4% | S&P 500 −5.2% | Correlation −0.50

2 Introduction

Crude oil is one of the most strategically important commodities in the global economy. As a primary input to production and transportation, oil price movements influence corporate earnings, inflation expectations, and equity market valuations. However, the nature of this relationship is far from constant — it shifts across economic regimes and is fundamentally disrupted by geopolitical shocks.

This project investigates how oil price shocks affect equity markets at both the index level and the individual stock level. We build a structured Python package to systematically measure these relationships from the 2020 pandemic to the most recent Iran conflict and extend the analysis to a cross-sectional view of 26 major S&P 500 constituents.

2.1 Research Questions

- What is the baseline statistical relationship between oil returns and S&P 500 returns?
- How does this relationship change during major geopolitical and economic shocks?
- Does the 2025–2026 Iran war represent a structural break in the oil–equity relationship?
- Can oil price shocks be systematically detected and quantified?
- How does oil sensitivity differ across sectors and individual stocks once broad market effects are controlled for?
- Does a supply-driven oil shock produce different cross-sectional return patterns than a demand-driven shock?

2.2 Scope & Data

- S&P 500 Index (SPX) — daily closing prices, December 2014 – March 2026
- Crude Oil Front-Month Futures (CL1) same date range, 2,828 trading days
- 26 individual S&P 500 stocks across six sectors (Part B) downloaded via Yahoo Finance
- VIX Volatility Index — used as a control variable in the multi-stock regressions
- Eight defined event windows spanning geopolitical crises, pandemics, and wars
- Data sources: Bloomberg Terminal (Part A); Yahoo Finance via yfinance (Part B)

3 Methodology

All analysis is conducted using a custom Python package, oil_spx_analyzer, comprising five modules. The package uses only standard scientific Python libraries — pandas, numpy, and matplotlib — with no external statistical packages such as statsmodels or seaborn.

3.1 Package Architecture

Module	File	Responsibility
Data Loader	data_loader.py	Reads Bloomberg Excel data, cleans dates, merges assets into a single DataFrame
Analyser	analyzer.py	Computers log returns, rolling correlation, volatility, OLS regression, shock detection

Module	File	Responsibility
Events	events.py	Defines eight event windows: computes per-event summary statistics
Report	report.py	Generates eight publication-ready charts and a formatted console report
Multi-Stock	multi_stock.py	Downloads 26 stocks via yfinance; runs multiple regression controlling for SPX and VIX; produces sector-level charts

3.2 Log Returns

Daily log returns are calculated for both assets to ensure stationarity and comparability: $r(t) = \ln(P(t) / P(t-1))$. Log returns are preferred over simple returns as they are time-additive and approximately normally distributed for large samples.

3.3 Rolling Correlation

A 30-day rolling Pearson correlation is used to capture the time-varying nature of the oil–equity relationship. A single full-sample figure would obscure the dramatic regime shifts that occur during crisis periods.

3.4 Simple OLS Regression (Part A Baseline)

A simple OLS regression is estimated across the full sample: $SPX(t) = \alpha + \beta \times Oil(t) + \varepsilon(t)$. The β coefficient measures the average S&P 500 return per unit move in oil. The model is implemented entirely in NumPy using the normal equations, without any external statistics libraries.

3.5 Multiple Regression Model (Part B)

The multi-stock extension estimates the following model for each of the 26 stocks:

$$\mathbf{Stock_i(t) = \alpha + \beta_1 \cdot Oil(t) + \beta_2 \cdot SPX(t) + \beta_3 \cdot VIX(t) + \varepsilon(t)}$$

Adding SPX and VIX as control regressors means that β_1 captures the pure oil sensitivity of each stock — the portion of its return attributable to oil after removing the influence of broad market direction and the volatility regime. This is a materially more informative measure than the simple beta from Part A.

3.6 Oil Price Shock Detection

A shock day is defined as any trading day where the absolute oil log return exceeds 3%. This threshold identifies extreme energy market moves and enables analysis of the same-day market response. Across the full sample, oil experienced a shock on 17.4% of all trading days — nearly one in five.

3.7 Stock Universe

The 26 stocks selected for Part B span six sectors chosen for their contrasting expected relationships with oil prices. The expected direction of β_1 is informed by economic theory:

Sector	Tickers	Expected β_1 Direction	Economic Rationale
Energy	XOM, CVX, COP, SLB, EOG	Strongly positive	Oil revenue directly drives earnings

Sector	Tickers	Expected β_1 Direction	Economic Rationale
Airlines	DAL, LUV, AAL, UAL	Negative	Jet fuel is 20–30% of operating cost
Technology	AAPL, MSFT, NVDA, GOOGL, META	Small negative	No direct commodity cost; negative β_1 driven by risk-off channel during supply shocks
Financials	JPM, GS, BAC, WFC	Mild positive	Macro growth channel
Consumer	AMZN, WMT, MCD	Slight negative	Input cost pass-through
Industrials	BA, CAT, GE	Moderate positive	Correlated with economic activity
Healthcare	JNJ, UNH	Low sensitivity	Healthcare is modestly affected by the energy cycle

4 Results & Analysis — Part A: Oil vs S&P 500

4.1 Full-Sample Price Behavior

Figure 1 shows the normalized price series for both assets, rebased to 100 at December 2014. Crude oil demonstrates substantially greater volatility than the S&P 500, with multiple severe drawdown episodes including the 2020 price war, the pandemic crash, and the 2025–2026 Iran conflict. The S&P 500 exhibits a long-run upward trend punctuated by sharp but recoverable drawdowns.

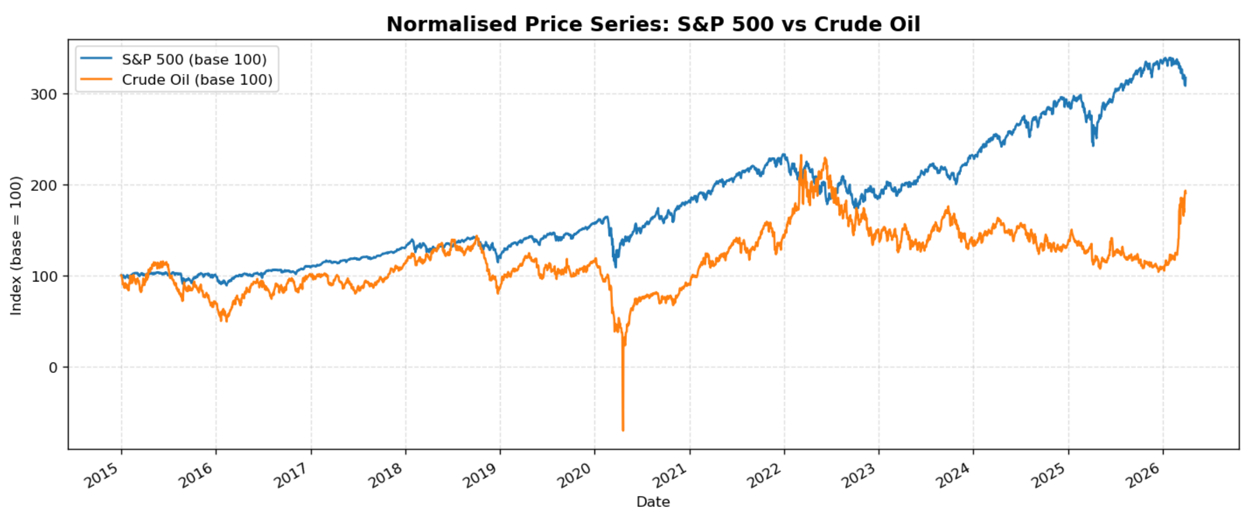

Figure 1: Normalized Price Series — S&P 500 vs Crude Oil (Base = 100, December 2014)

4.2 Volatility Comparison

Asset	Annualized Volatility	Interpretation
S&P 500 (SPX)	17.86%	Typical for a broad diversified equity index
Crude Oil (CL1)	45.70%	More than 2.5× the volatility of equities

4.3 Log Returns & Volatility Clustering

Figure 2 shows daily log returns for both assets. Both series exhibit clear volatility clustering — calm periods interrupted by sharp bursts of extreme moves. The largest spikes coincide with the event windows analyzed in Section 4.6. Crude oil returns reach magnitudes of ±25–30% on individual days during the most acute crises, compared to a maximum of approximately ±12% for the S&P 500.

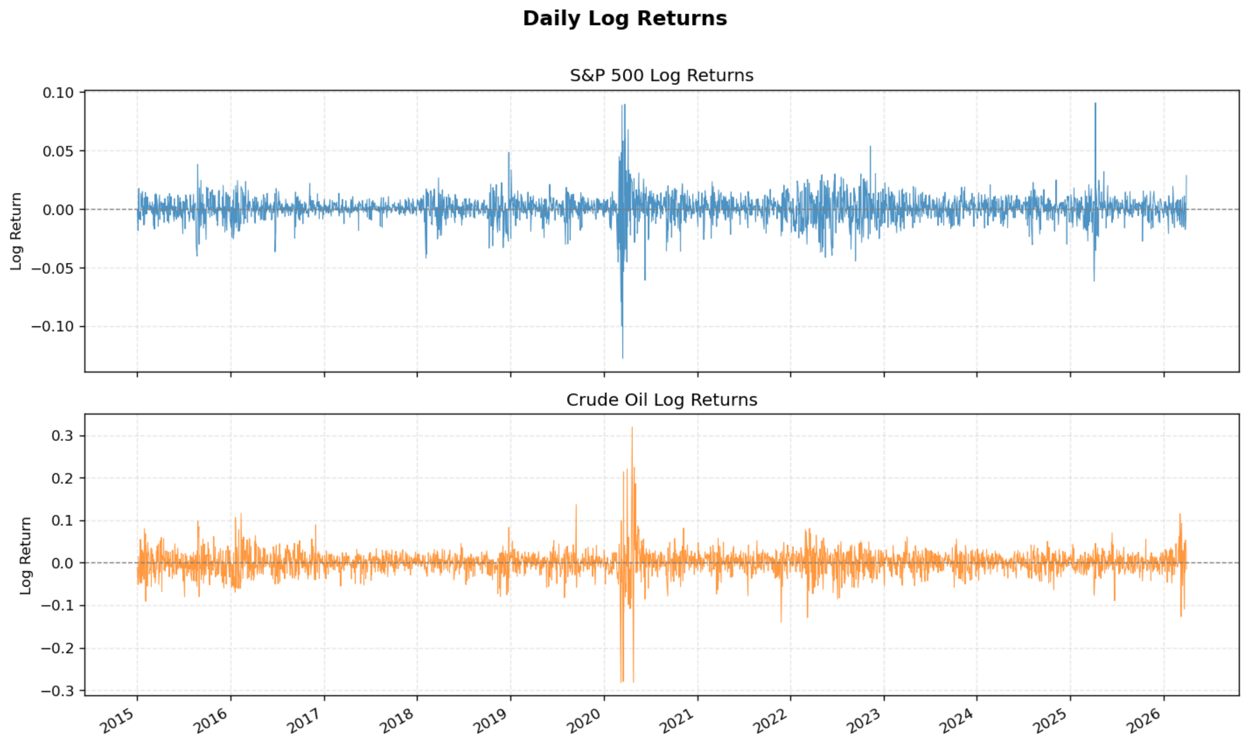

Figure 2: Daily Log Returns — S&P 500 (top panel) and Crude Oil (bottom panel)

4.4 Rolling Correlation Analysis

Figure 3 shows the 30-day rolling correlation between oil and S&P 500 returns across the full sample. The correlation fluctuates substantially — from strongly positive to strongly negative — confirming that a single static figure is insufficient to characterize this relationship. During normal economic expansion, the two assets exhibit mild positive correlation consistent with the growth narrative. During geopolitical supply shocks, the correlation turns sharply negative.

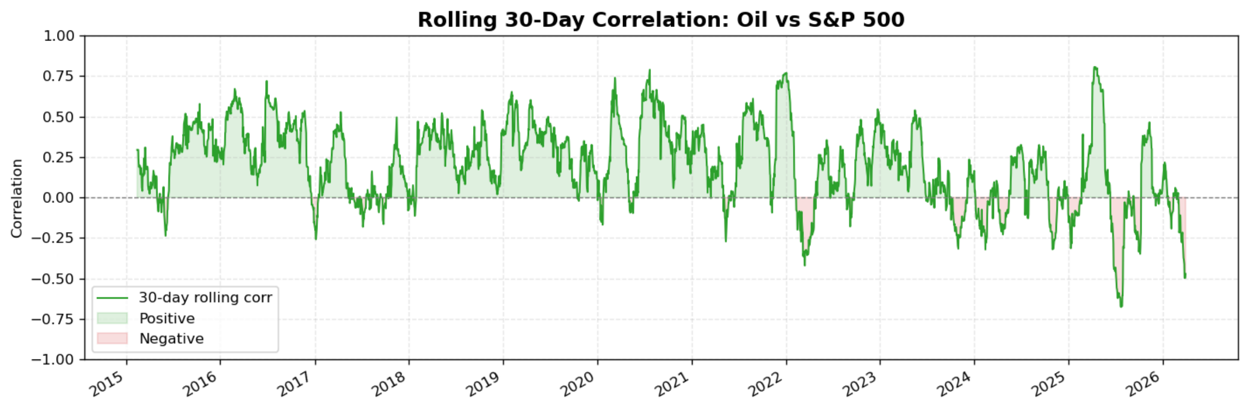

Figure 3: 30-Day Rolling Correlation — Oil vs S&P 500

4.5 OLS Regression Results

The full-sample OLS regression quantifies the average sensitivity of S&P 500 returns to oil price movements. Results are presented in Table 2 below.

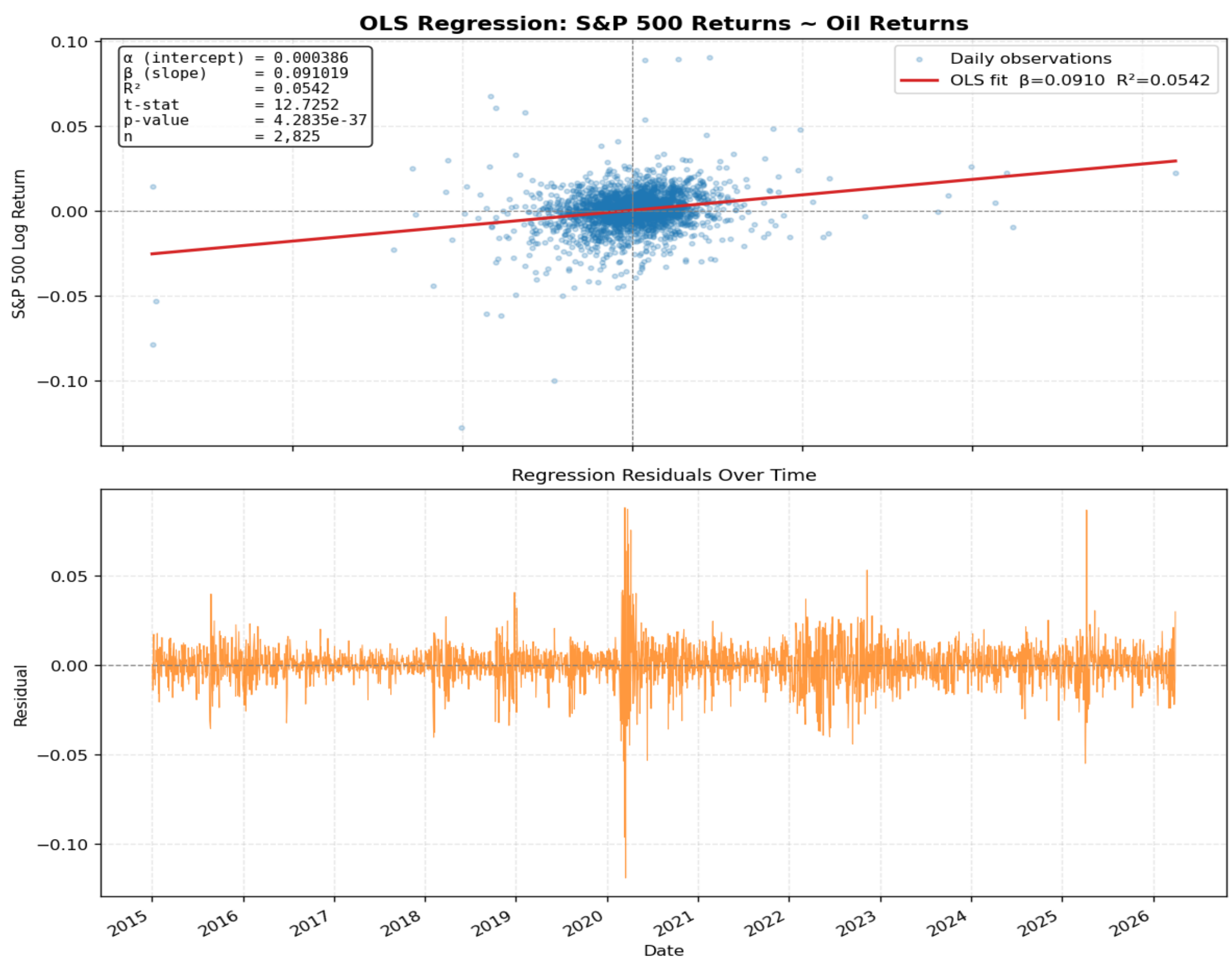

Table 2: Full-Sample OLS Regression Results — SPX(t) = α + β × Oil(t) + ε(t)

Statistic	Value	Interpretation
α (Intercept)	0.000386	Small positive daily drift, independent of oil
β (Slope)	0.091019	A 1% oil move → 0.091% S&P 500 move on average
R^2	0.0542	Oil explains 5.4% of daily S&P 500 variation
t-statistic	12.7252	Highly statistically significant
p-value	4.28×10^{-37}	Relationship is not attributable to chance
Observations	2,828	Full sample of available trading days

Figure 4: OLS Regression — Oil vs S&P 500 log returns with fitted line (top); residuals over time (bottom)

Key Finding

While β is highly significant ($p < 0.001$), the R^2 of 0.054 shows that oil alone is a poor predictor of daily equity returns. Many forces — monetary policy, earnings, sentiment — dominate normal trading. However, during geopolitical crises oil becomes a dominant driver, and the direction of the relationship reverses. Part B addresses this by controlling market-wide effects and analyzing individual stocks.

4.6 Event Window Analysis

Eight distinct event windows are analyzed, measuring cumulative returns and within-event oil–SPX correlation. Figure 5 shows normalized price behavior during each individual event window. Table 3 summarizes all eight events.

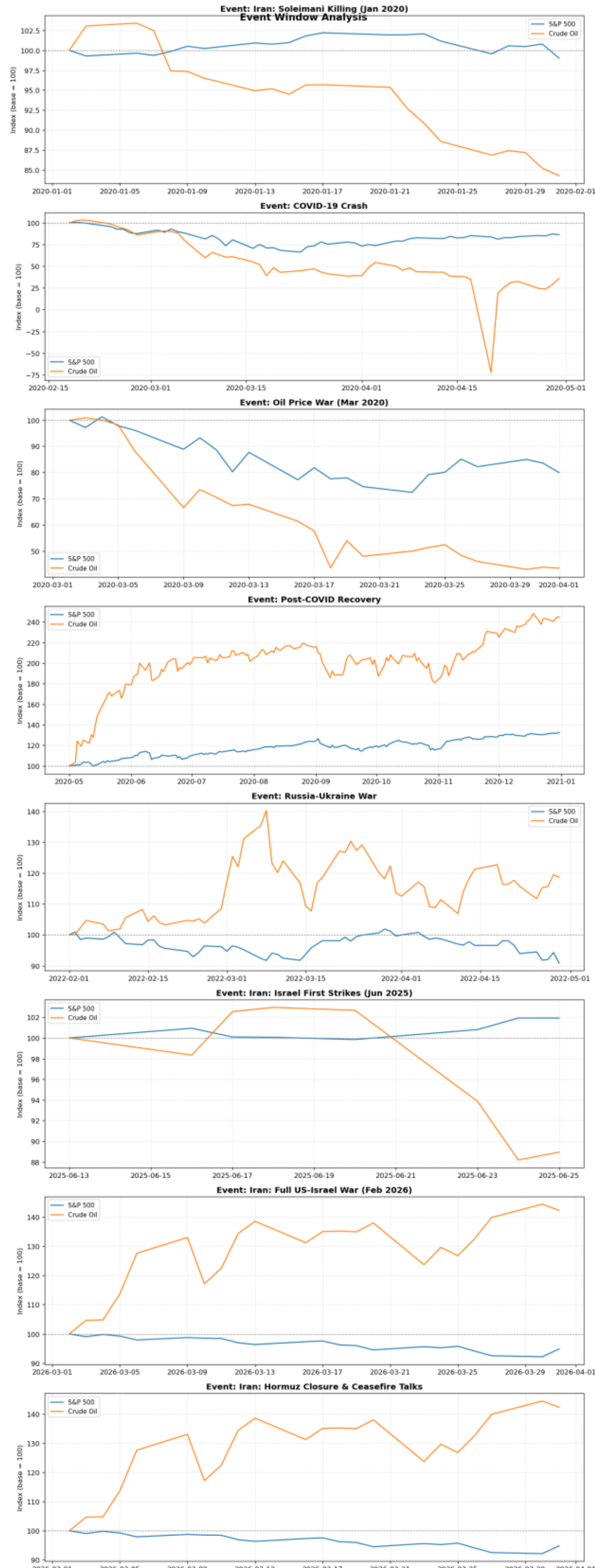
Event: Iran: Soleimani Killing (Jan 2020)
Event Window Analysis
S&P 500
Crude Oil
Index (base = 100)
Event: COVID-19 Crash
Event: Oil Price War (Mar 2020)
Event: Post-COVID Recovery
Event: Russia-Ukraine War
Event: Iran: Israel First Strikes (Jun 2025)
Event: Iran: Full US-Israel War (Feb 2026)
Event: Iran: Hormuz Closure & Ceasefire Talks

Table 3: Event Impact Summary — All Eight Event Windows

Event	Period	Days	Oil Cum. Return	SPX Cum. Return	Correlation
Iran: Soleimani Killing	Jan 2020	21	−16.9%	−0.2%	+0.11
COVID-19 Crash	Feb–Apr 2020	50	−41.5%	−9.97%	+0.27
Oil Price War	Mar–Apr 2020	23	−79.0%	−17.9%	+0.43
Post-COVID Recovery	May–Dec 2020	170	+94.6%	+25.4%	+0.38
Russia–Ukraine War	Feb–Apr 2022	62	+17.2%	−8.9%	−0.18
Iran: Israel Strikes	Jun 2025	8	−4.7%	+0.8%	−0.92
Iran: Full US-Israel War	Feb–Mar 2026	22	+41.4%	−5.2%	−0.50
Iran: Hormuz Closure	Mar–Apr 2026	22	+41.4%	−5.2%	−0.50

Note: Iran: Israel Strikes ran only 8 trading days; the initial risk premium partially reversed, producing a net oil decline. The −0.92 correlation is the highest magnitude negative value in the full dataset. The Hormuz Closure period (Mar 1 – Apr 7, 2026) overlaps with and extends beyond the dataset's March 2026 cutoff, so both events are truncated to the same 22 available trading days — hence the identical figures. The Hormuz Closure row reflects the combined shock period rather than an independent event window.

Figure 5: Normalized Price Performance During Each Event Window

Figure 6 provides a visual comparison of cumulative returns across all events.

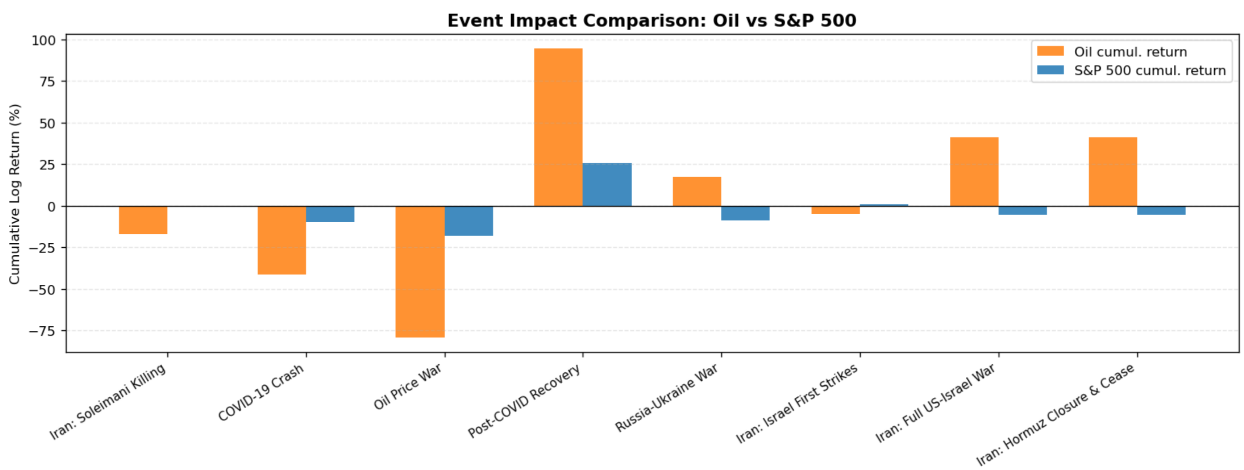

Figure 6: Cumulative Return Comparison Across All Events — Oil (orange) vs S&P 500 (blue)

5 Iran War 2025–2026: Deep Dive

The 2025–2026 Iran conflict is the most significant geopolitical shock in our dataset. Unlike previous crises, it unfolded in two distinct phases — an initial Israeli strike campaign in June 2025 and a full US-Israel military operation beginning February 2026 — each producing different oil and equity market reactions.

5.1 Timeline of Key Events

Date	Event	Oil Market Impact
Jun 13, 2025	Israel launches first airstrikes on Iran nuclear facilities	Oil +7% on open; $10–15/bbl geopolitical risk premium added

Date	Event	Oil Market Impact
Jun 25, 2025	12-day campaign ends; supply disruption limited	Risk premium fades; oil partially retraces
Feb 28, 2026	Full US-Israel strike campaign on Iran begins	Oil surges 10–13% on first trading day
Mar 9, 2026	Brent crude crosses $100 per barrel	First time above $100 since the 2022 Russia-Ukraine spike
Mar 2026	Strait of Hormuz effectively closed	~20% of global seaborne oil supply disrupted
Apr 7, 2026	Ceasefire talks commence	Oil volatility remains elevated; markets cautious

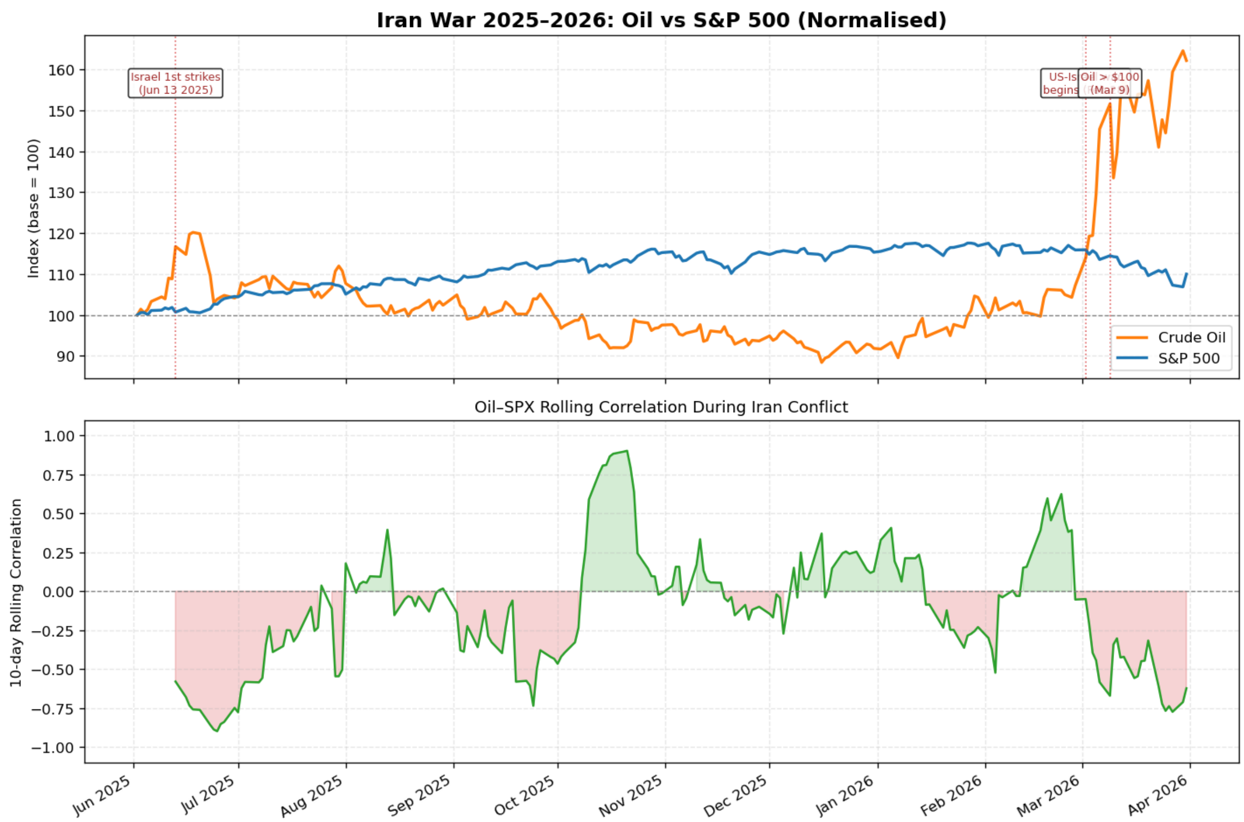

Table 4: All-Stock Returns — Iran: Full US-Israel War (Feb–Mar 2026, 22 trading days)

Ticker	Sector	Cum. Return
LUV	Airlines	−30.1%
AAL	Airlines	−25.0%
GE	Industrials	−22.4%
UAL	Airlines	−22.1%
META	Technology	−18.9%
BA	Industrials	−18.4%
GOOGL	Technology	−13.0%
CAT	Industrials	−10.7%
UNH	Healthcare	−10.6%

MCD	Consumer	−9.5%
MSFT	Technology	−9.0%
NVDA	Technology	−7.0%
AAPL	Technology	−6.9%
WFC	Financials	−5.9%
GS	Financials	−5.7%
JPM	Financials	−5.7%
BAC	Financials	−4.8%
AMZN	Consumer	−4.4%
DAL	Airlines	−3.9%
WMT	Consumer	−3.3%
JNJ	Healthcare	−2.4%
SLB	Energy	+0.4%
XOM	Energy	+11.7%
CVX	Energy	+12.1%
COP	Energy	+15.8%
EOG	Energy	+18.9%

Figure 7: Iran War 2025–2026 — Normalized Oil vs S&P 500 with event markers and 10-day rolling correlation

5.2 Key Findings

- During the June 2025 Israeli strikes, the within-event daily correlation reached −0.92 — the strongest negative value in the entire 11-year dataset. This measures day-by-day co-movement: in the initial shock days oil spiked sharply, and equities fell, then both reversed as the risk premium faded. The −0.92 reflects the tightness of that daily inverse relationship, not the net cumulative returns over the window.
- During the February 2026 full US-Israel war, oil rose +41.4% cumulatively while the S&P 500 fell −5.2%, producing a clear negative correlation of −0.50.
- Oil crossed $100 per barrel on March 9, 2026 — for the first time since the post-Ukraine spike of 2022.
- The Strait of Hormuz closure disrupted approximately 20% of global seaborne oil supply, the most severe energy security event since the 1970s.

Key Finding

The Iran 2025–2026 conflict is the clearest demonstration in our dataset that supply-driven oil shocks reverse the normal oil–equity relationship. When oil rises due to geopolitical fear rather than economic demand, equity markets fall simultaneously — the opposite of the normal cycle. The multi-stock extension in Part B shows this effect is far larger for Energy stocks (who benefit) and Airlines (who are harmed) than the index-level figure suggests.

6 Oil Price Shock Analysis

A shock day is defined as any trading day where the absolute crude oil return exceeds 3%. This threshold captures extreme oil price movements and allows systematic analysis of the concurrent equity market response. The 3% threshold corresponds approximately to a 2-standard-deviation daily move given crude oil's annualized volatility of 45.7% (daily $\sigma \approx 2.87\%$), ensuring only genuinely extreme sessions are classified while retaining sufficient events (491 shock days) for meaningful statistical analysis.

6.1 Shock Statistics

Metric	Value
Total shock days (|oil return| > 3%)	491
Positive shocks (oil price spike)	229 (46.6%)
Negative shocks (oil price crash)	262 (53.4%)
Overall shock frequency	17.4% of all 2,828 trading days

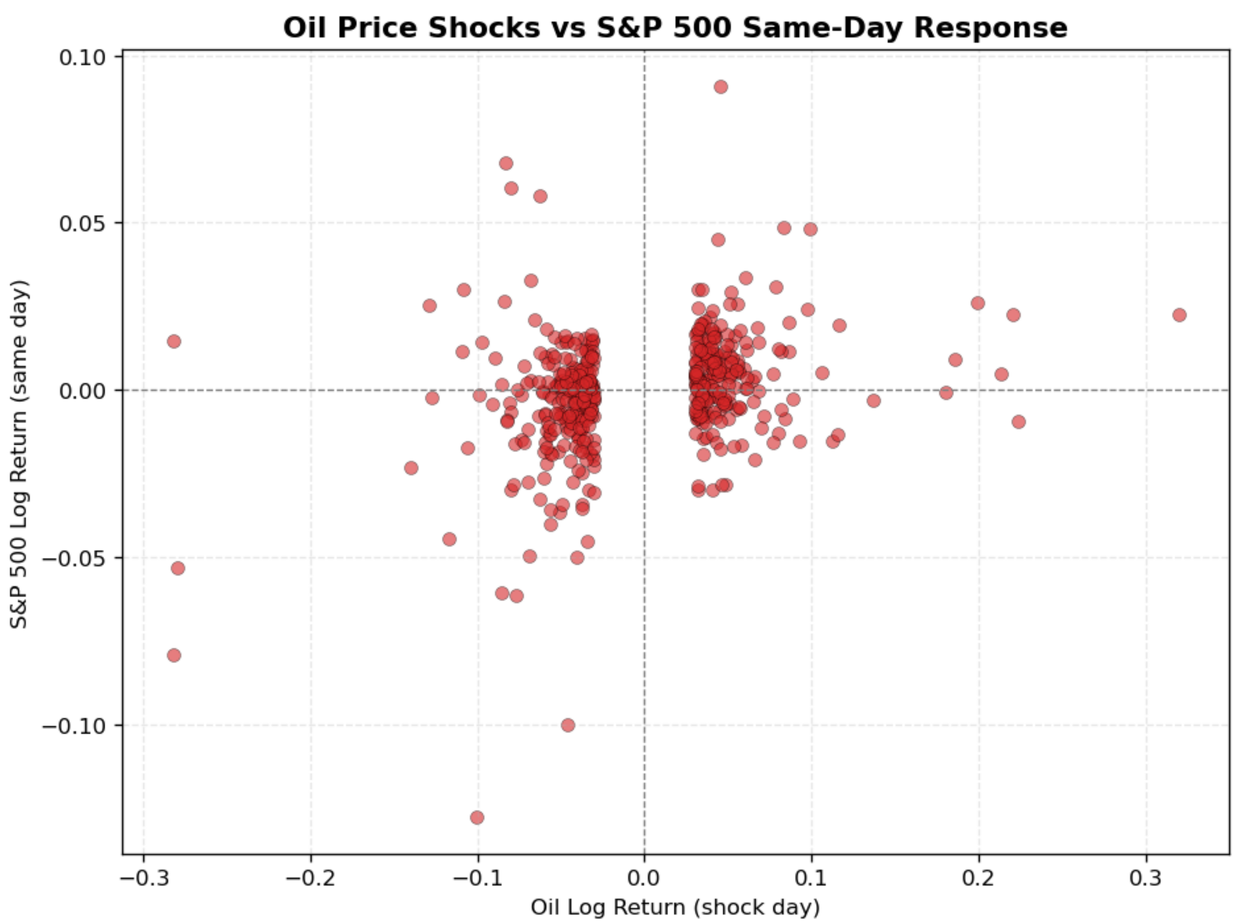

Figure 8: Oil Shock Days — Oil log return vs same-day S&P 500 return

Figure 8 reveals an important asymmetry. During positive oil shocks (right half), S&P 500 returns are broadly dispersed with a slight positive skew, suggesting demand-driven oil spikes do not consistently harm equities. During negative oil shocks (left half), S&P 500 returns cluster more negatively — oil crashes driven by recession fears tend to pull equities down simultaneously.

7 Results & Analysis — Part B: Multi-Stock Multiple Regression

Part B extends the analysis from the broad S&P 500 index to 26 individual constituents across six sectors. The multiple regression model controls for broad market returns (SPX) and the volatility regime (VIX), isolating β_1 as the pure oil sensitivity of each stock.

7.1 Multiple Regression Results — All 26 Stocks

Table 5 presents the key regression output for all 26 stocks. The β_1 (oil) coefficient is the primary variable of interest. The t-statistic and significance level indicate whether the pure oil effect is statistically distinguishable from zero once market-wide movements are removed.

Sector	Ticker	β_1 Oil	β_2 SPX	Adj R^2	Sig.
Airlines	AAL	−0.0958	1.4072	0.255	***
Airlines	LUV	−0.0648	1.0308	0.274	***
Airlines	UAL	−0.0517	1.4269	0.279	**
Airlines	DAL	−0.0481	1.2802	0.330	**
Consumer	WMT	−0.0302	0.5529	0.182	***
Consumer	AMZN	−0.0281	1.1049	0.409	*
Consumer	MCD	−0.0072	0.7194	0.320	
Energy	XOM	+0.2423	0.7050	0.436	***
Energy	CVX	+0.2438	0.9299	0.481	***
Energy	SLB	+0.3404	0.9607	0.413	***
Energy	COP	+0.3895	0.9258	0.482	***
Energy	EOG	+0.3973	0.8194	0.429	***
Financials	WFC	+0.0108	1.1649	0.432	
Financials	GS	+0.0252	1.1359	0.549	**
Financials	JPM	+0.0278	1.0823	0.525	***
Financials	BAC	+0.0311	1.1920	0.496	**
Healthcare	JNJ	−0.0211	0.5573	0.232	**
Healthcare	UNH	−0.0100	0.8830	0.228	
Industrials	GE	+0.0228	1.1136	0.331	
Industrials	BA	+0.0374	1.4306	0.366	*
Industrials	CAT	+0.0887	0.9318	0.427	***
Technology	NVDA	−0.0444	1.6529	0.433	**
Technology	META	−0.0370	1.3188	0.373	**
Technology	MSFT	−0.0237	1.2018	0.609	**
Technology	GOOGL	−0.0216	1.1036	0.501	*
Technology	AAPL	−0.0176	1.2189	0.554	

*All β values from 2,828 daily observations (Dec 2014–Mar 2026) via NumPy normal equations controlling for SPX and VIX. β_3 (VIX) omitted for brevity. Significance: *** $p<0.001$ ** $p<0.01$ * $p<0.05$.*

7.2 Oil Beta Ranking

Figure 9 ranks all 26 stocks from most negatively oil-sensitive (Airlines, left) to most positive (Energy, right). The color coding by sector makes the sector gradient immediately visible — confirming that the model is capturing real economic relationships rather than statistical noise.

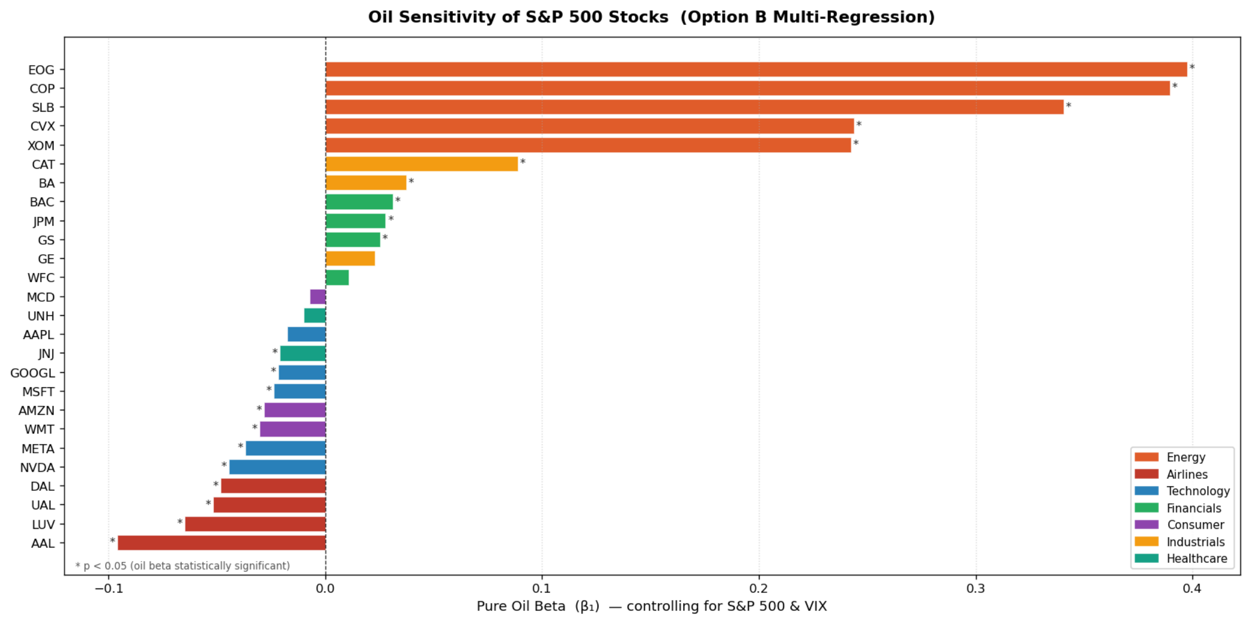

Figure 9: Oil Beta Ranking — All 26 Stocks Sorted by Pure Oil Sensitivity (β_1), Colour-Coded by Sector

7.3 Sector Heatmap

Figure 10 presents a two-panel heatmap. The left panel shows β_1 (pure oil sensitivity) for each stock — red indicates negative sensitivity, green indicates positive. The right panel shows the adjusted R^2 of the full three-regressor model, indicating how well Oil, SPX, and VIX together explain each stock's daily returns.

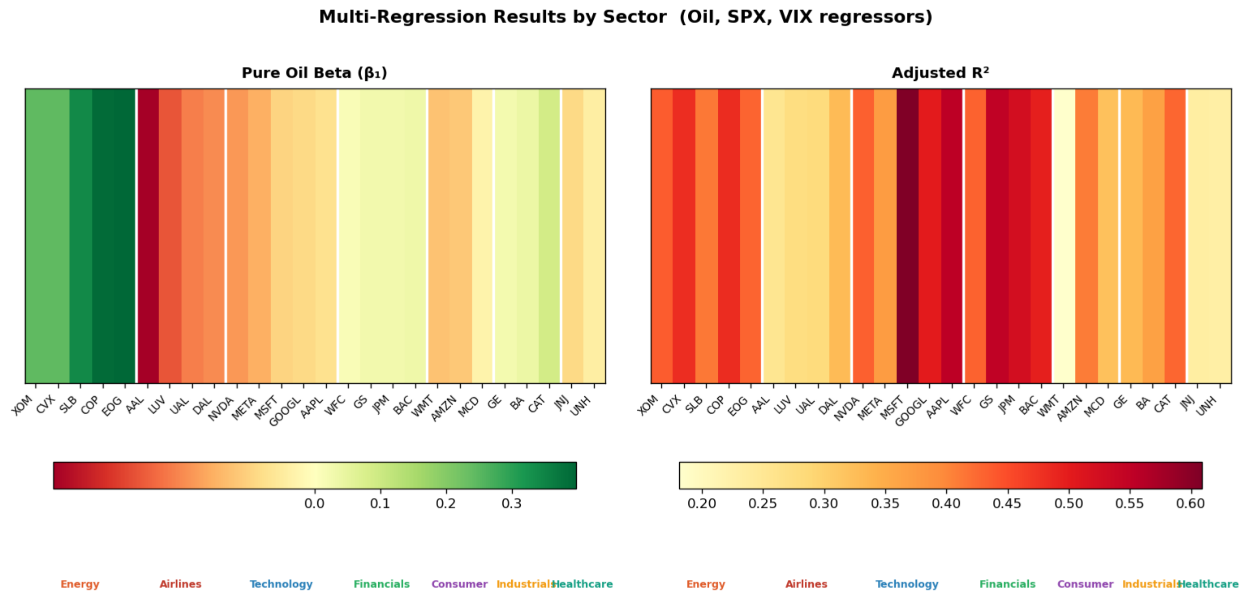

Figure 10: Sector Heatmap — Pure Oil Beta β_1 (left panel) and Adjusted R^2 (right panel)

7.4 Rolling 60-Day Pure Oil Beta

Figure 11 shows the rolling 60-day pure oil beta for four representative stocks — XOM (Energy), DAL (Airlines), AAPL (Technology), and JPM (Financials). Each panel uses the same multi-regressor OLS on each rolling window, so the plotted beta is always the component of return attributable to oil alone, net of market and volatility effects.

The most striking feature is the divergence during the Iran war periods: XOM's beta spikes sharply positive while DAL's beta deepens negative — the opposite ends of the spectrum responding in opposite directions to the same shock.

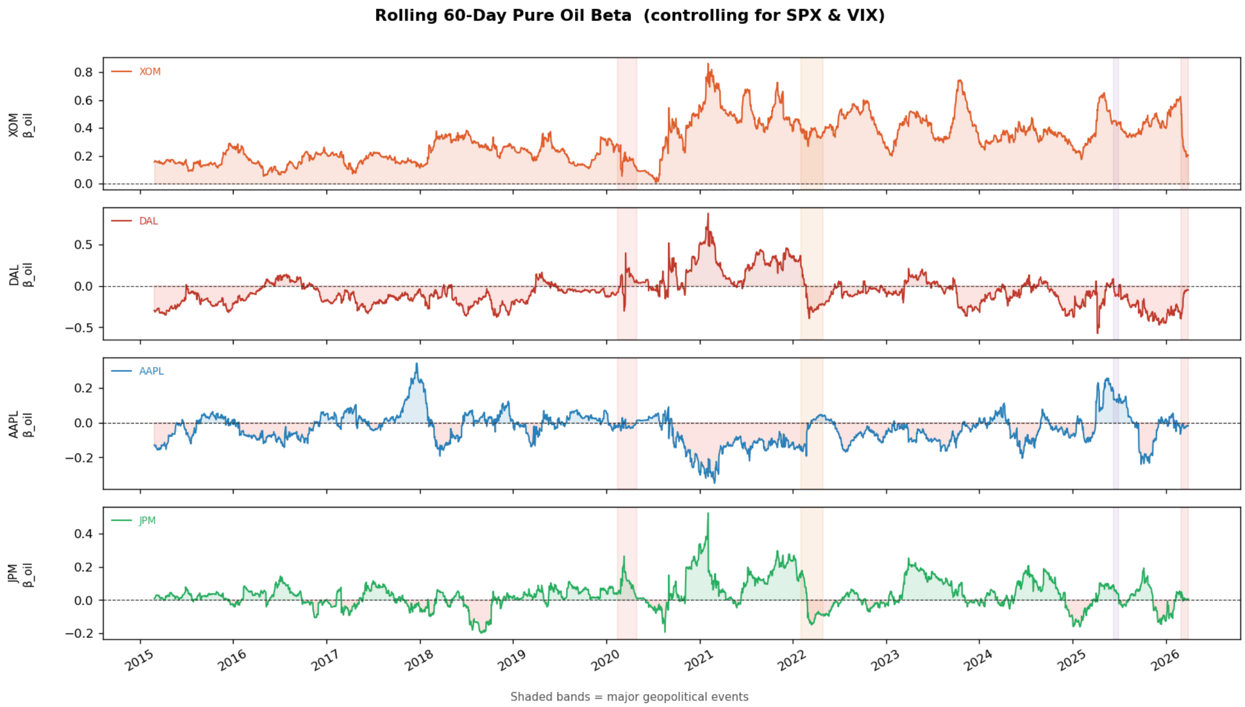

Figure 11: Rolling 60-Day Pure Oil Beta — XOM, DAL, AAPL, JPM (shaded bands = geopolitical events)

7.5 Event Heatmap — All Stocks × All Events

Figure 12 is the centerpiece chart of Part B. Rows corresponds to the eight event windows from Part A; columns correspond to the 26 stocks sorted left-to-right by full-sample oil beta (most negative to most positive). Green cells indicate positive cumulative returns during the event; red cells indicate negative returns.

The key observation is the contrasting pattern between event types. During the COVID Crash and Oil Price War, virtually the entire matrix is red — a demand-driven collapse in which no sector escaped. During the Post-COVID Recovery, the matrix is uniformly green. During the Iran War 2025–26, a clear left-to-right gradient emerges — Airlines (left columns) are red while Energy stocks (right columns) are green. This gradient is visible precisely because the columns are sorted by oil beta.

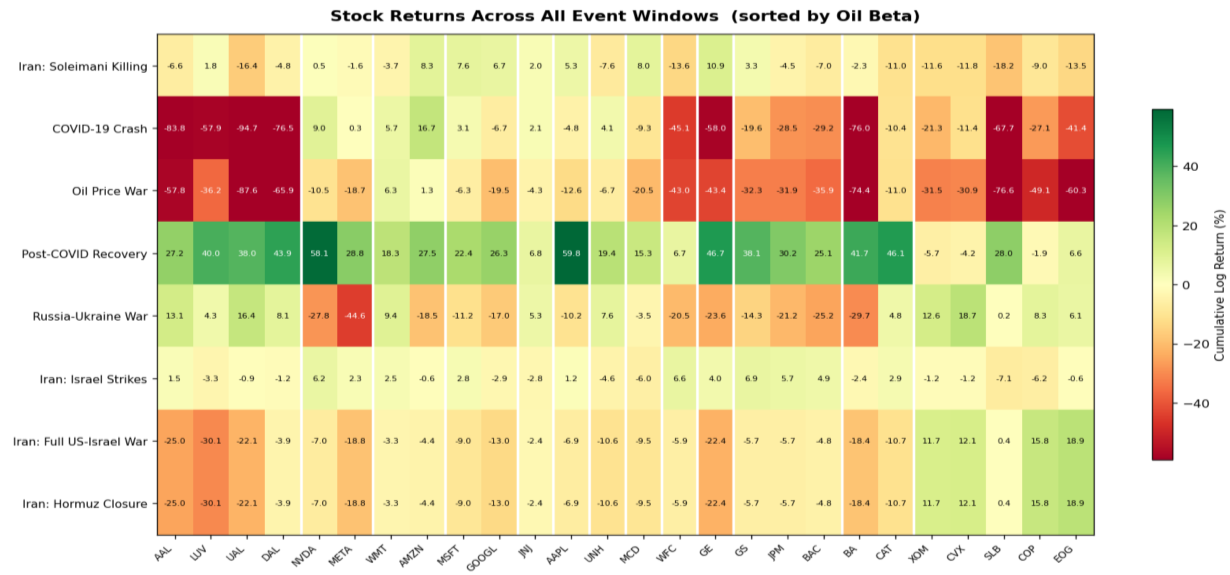

Figure 12: Event Heatmap — Cumulative Returns Across All Events and All Stocks (sorted by oil beta)

7.6 Model Validation: Beta vs Actual Return

Figures 13 and 14 test whether the full-sample pure oil beta predicts actual cross-sectional returns during specific events. Each point represents one stock; the x-axis shows its full-sample β_1 and the y-axis shows its cumulative return during the event. A positive slope on the regression line would indicate that the beta was predictive.

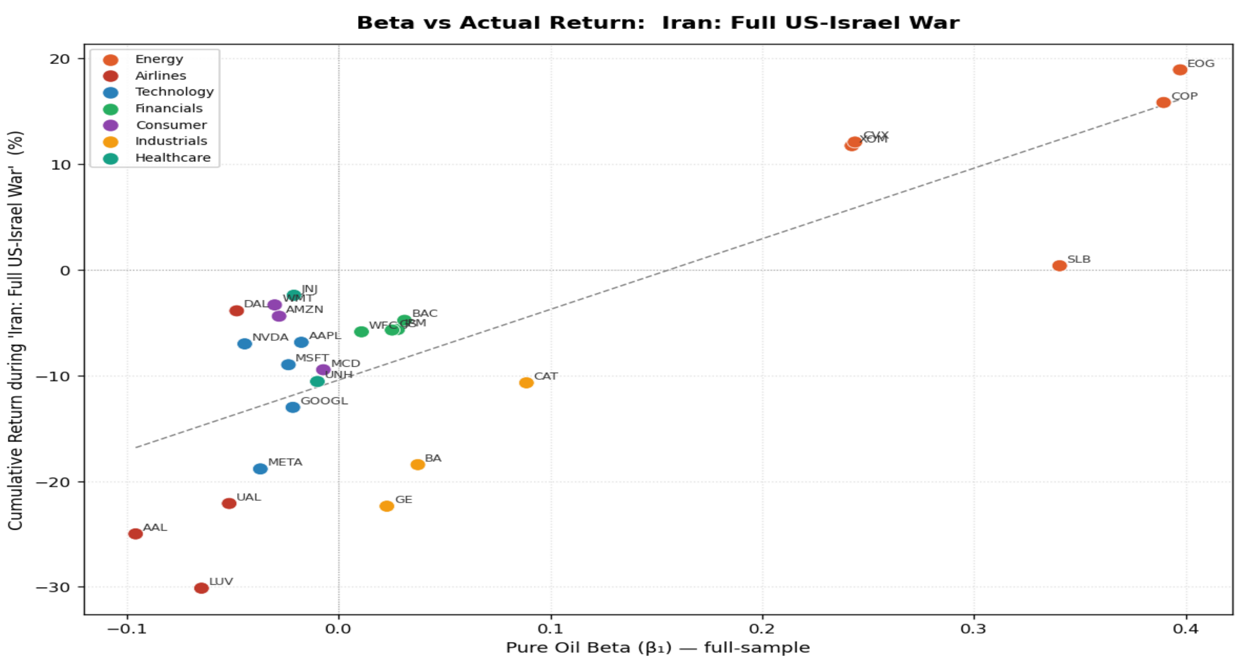

Figure 13: Beta vs Actual Return — Iran: Full US-Israel War. A positive slope indicates beta was predictive.

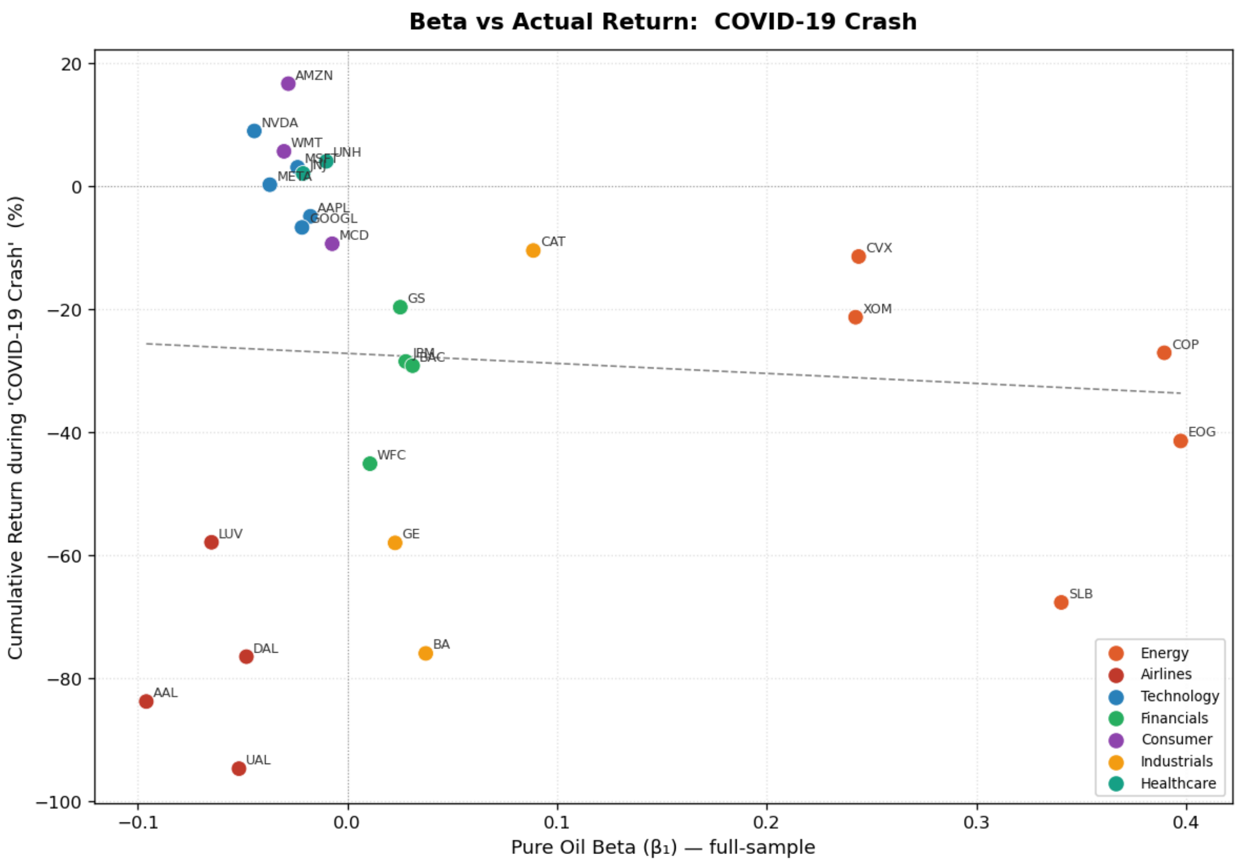

Figure 14: Beta vs Actual Return — COVID-19 Crash. A flat slope indicates beta was not predictive.

The contrast between Figure 13 (Iran War, supply shock) and Figure 14 (COVID, demand shock) is striking. During the Iran War, the regression line has a clearly positive slope — Energy stocks with high β_1 outperformed and Airlines with negative β_1 underperformed, exactly as predicted. During the COVID crash, the slope is near-zero — all stocks fell together regardless of their oil sensitivity, confirming that demand-driven oil shocks operate through a different mechanism.

7.7 Individual Stock Deep-Dive

Four representative stocks — XOM (Energy), DAL (Airlines), NVDA (Technology), and JPM (Financials) are examined in full multi-regression detail via the notebook's Part B §11. The β_1 coefficients confirm the sector narrative from Section 7.1, while β_2 and β_3 reveal how strongly each stock co-moves with the broad market and the volatility regime.

XOM (Energy). $\beta_1 = +0.242$ ($p < 0.001$). XOM's pure oil sensitivity is strongly positive and stable. Its $\beta_2 \approx 0.71$ indicates below-market-beta for a large-cap stock, consistent with energy sector dynamics. The rolling beta spikes sharply during both Iran war phases — the clearest single-stock illustration of the supply-shock mechanism in the dataset. Adjusted $R^2 \approx 0.44$.

DAL (Airlines). $\beta_1 = -0.048$ ($p < 0.01$). DAL's negative oil sensitivity reflects jet fuel's 20–30% share of operating costs. Its $\beta_2 \approx 1.28$ means Delta amplifies broad market swings. Controlling for SPX is essential here: the naïve simple regression would substantially overstate the oil effect by conflating it with market-wide co-movement. Adjusted $R^2 \approx 0.33$.

NVDA (Technology). $\beta_1 = -0.044$ ($p < 0.01$). Statistically significant but economically small. NVDA's dominant driver is its very high market sensitivity ($\beta_2 \approx 1.65$) — NVIDIA amplifies broad risk-on/risk-off moves far more than it responds to oil. The negative oil sign likely reflects that oil spikes (geopolitical) coincide with equity risk-off, and NVDA as a high-beta growth stock falls disproportionately in such episodes. Adjusted $R^2 \approx 0.43$.

JPM (Financials). $\beta_1 = +0.028$ ($p < 0.001$). JPM's mild positive oil sensitivity reflects the macro growth channel: oil strength tends to accompany economic expansion and credit growth that benefits bank revenues. With $\beta_2 \approx 1.08$ and adjusted $R^2 \approx 0.53$ — the highest outside the Energy sector — SPX and VIX together explain more than half of JPMorgan's daily variation, confirming its role as a bellwether for broad economic conditions.

8 Conclusions

8.1 Part A — Oil vs S&P 500

Finding	Detail
Baseline relationship is statistically significant but weak	Oil explains only 5.4% of daily S&P 500 variation ($R^2 = 0.054$)
β is highly significant	$t = 12.73$, $p = 4.28 \times 10^{-37}$ — the relationship is not random
Relationship is not stable	30-day rolling correlation swings from −0.92 to +0.80 across the sample
Geopolitical shocks reverse the sign	Iran War 2025–26: oil +41.4%, SPX −5.2%, correlation −0.50
Shock frequency is high	Oil moved more than 3% on 17.4% of all trading days

8.2 Part B — Multi-Stock Multiple Regression

Finding	Detail
Sector heterogeneity is large	Energy β_{oil}: +0.24 to +0.40 (avg ≈ +0.31). Airlines β_{oil}: −0.05 to −0.10 (avg ≈ −0.07). The same shock drives opposite outcomes
Pure β_{oil} differs from naive β	Controlling for SPX and VIX materially changes oil coefficients for Financials and Industrials
Iran War validates the model cross-sectionally	Stocks with high full-sample β_{oil} outperformed; low-β_{oil} stocks underperformed
COVID does not validate it	During the demand crash, β_{oil} was not predictive — all sectors fell together
Technology stocks appear relatively insulated from oil-price movements.	Technology stocks (AAPL, MSFT, GOOGL, META) show small negative β_{oil}; NVDA and MSFT are statistically significant ($p<0.01$) but economically small

8.3 Central Conclusion

Key Finding

The oil–equity relationship is dynamic, regime-dependent, and sector-specific. A supply-driven oil shock (geopolitical) and a demand-driven shock (recession or pandemic) produce opposite cross-sectional return patterns across S&P 500 stocks. The direction of oil's causality — not just its magnitude — determines whether Energy stocks outperform Airlines or whether both move together. A portfolio manager monitoring only the index-level oil–equity correlation would miss the sector-level divergence that becomes visible in the multi-stock analysis.

9 References & Data Sources

Data Sources

- Bloomberg Terminal — S&P 500 Index (SPX) daily closing prices, December 2014 – March 2026
- Bloomberg Terminal — Crude Oil Front-Month Futures (CL1) daily closing prices, December 2014 – March 2026
- Yahoo Finance via yfinance — 26 individual S&P 500 stock prices (Part B)
- Yahoo Finance via yfinance — VIX Volatility Index (^VIX) daily values

Geopolitical Event References

- Goldman Sachs Global Investment Research — Iran Conflict Oil Market Impact Analysis, June 2025
- J.P. Morgan Markets — Middle East Escalation Energy Price Forecast, February 2026
- International Energy Agency (IEA) Global Energy Security Emergency Assessment, March 2026
- U.S. Energy Information Administration (EIA) — Strait of Hormuz Oil Flow Disruption Impact Report

Technical References

- Python Software Foundation — Python 3.x programming language
- NumPy Development Team — NumPy: Array computing for Python (OLS regression implementation)
- Pandas Development Team — Pandas: Data manipulation and time series analysis library
- Matplotlib Development Team — Matplotlib: Visualization library for Python (all chart generation)
- Yahoo Finance / yfinance — Open-Source Python library for financial data download